1 Warm up (Estonian)

Problem 1 (2024-10-4). Pieces of cardboard of dimensions 1×4 are places on a 10×10 grid in such a way that each piece covers exactly 4 adjacent unit squares (either horizontally or vertically) and no two pieces touch each other side-to-side, edge-to-edge, or corner-to-corner. Find the largest possible number of cardboard pieces.

Problem 2 (2022-winter_open-S6). Andres, Bearu and Croot jointly count natural numbers 1 through 2022 in increasing order. Each of the must say exactly 674 numbers, but none of them may say two consecutive numbers. Prove that they have at least 10^{400} possibilities to perform counting according to the rules. Try to find a better bound.

Problem 3 (2024-VV-P6). Fix a natural number n. A function $f : \{0, 1, \ldots, n\} \to \{0, 1, \ldots, n\}$ is called regular if $f(0) = 0$ and $f(i) \in \{i-1, f(i-1), f(f(i-1)), \ldots\}$ for every $i = 1, \ldots n$. If, for instance, $n = 3$, then the function $f(0) = f(1) = 0, f(2) = f(3) = 1$ is regular, but the function $f(0) = f(1) = f(2) = 0, f(3) = 1$ is not. Juku chooses a regular function f and tells Miku for each $k = 0, 1, \ldots, n$, how many different arguments i are there such that $f(i) = k$. Can Miku always determine based on this information which regular function Juku had chosen?

Problem 4 (Swedish 1985 q1). Find the smallest positive integer n such that if the first digit is moved to become the last digit, then the new number is $7n/2$.

2 Folklore

Problem 5 (HMMT Team 2019/8). Can the set of lattice points $\{(x, y) \mid x, y \in \mathbb{Z}, 1 \leq x, y \leq 252, x \neq y\}$ be colored using 10 distinct colors such that for all $a \neq b$, $b \neq c$, the colors of (a, b) and (b, c) are distinct?

Problem 6 (Gray codes). Consider all bitstrings of length n (2^n of them). You wish to put them in a circle, so that the XOR of any two adjacent bitstrings has a unique 1. Find an explicit construction achieving this.

Problem 7 (IMSC 2023 combi mock, de Brujin sequence). An international conference consists of 2023 representatives from each of 2023 different countries. Prove that 2023^2 people can be seated around a large round table such that if A and B are two distinct representatives from the same country,

Problem 8 (ToT Spring 2016). A designer took a wooden cube $5 \times 5 \times 5$, divided each face into unit squares and painted each square black, white or red so that any two squares with a common side have different colours. What is the least possible number of black squares? (Squares with a common side may belong to the same face of the cube or to two different faces.)

Problem 9 (Spanish MO 2002 P3). The function g is defined on positive integers and satisfies the following conditions:

$$g(2) = 1 \quad g(2n) = g(n) \quad g(2n+1) = g(2n) + 1.$$

where n is a natural number such that $1 \leq n \leq 2002$.

Find the maximum value M of $g(n)$. Also, calculate how many values of n satisfy the condition of $g(n) = M$.